FEVER: results under the A2 attack

Table 1
Comparison with published methods under A2.

Method	Benign flows harmed (%)	Average attack share (%)	Maximum attack share (%)	Average flow table usage (%)	First packets lost (%)	New connections succeeding (%)	Attackers blocked	Decision time (ms)
None	10.29	58.4	94.8	75.2	21.6	30.3	0 / 10	29
FTMaster	**0.01**	**10.0**	58.1	**30.2**	**9.9**	27.2	0 / 10	**10**
DMS-BTT	0.02	18.7	**54.3**	38.5	20.0	31.0	0 / 10	28
FEVER	1.05	12.6	84.7	31.7	14.2	**47.4**	10 / 10	**10**

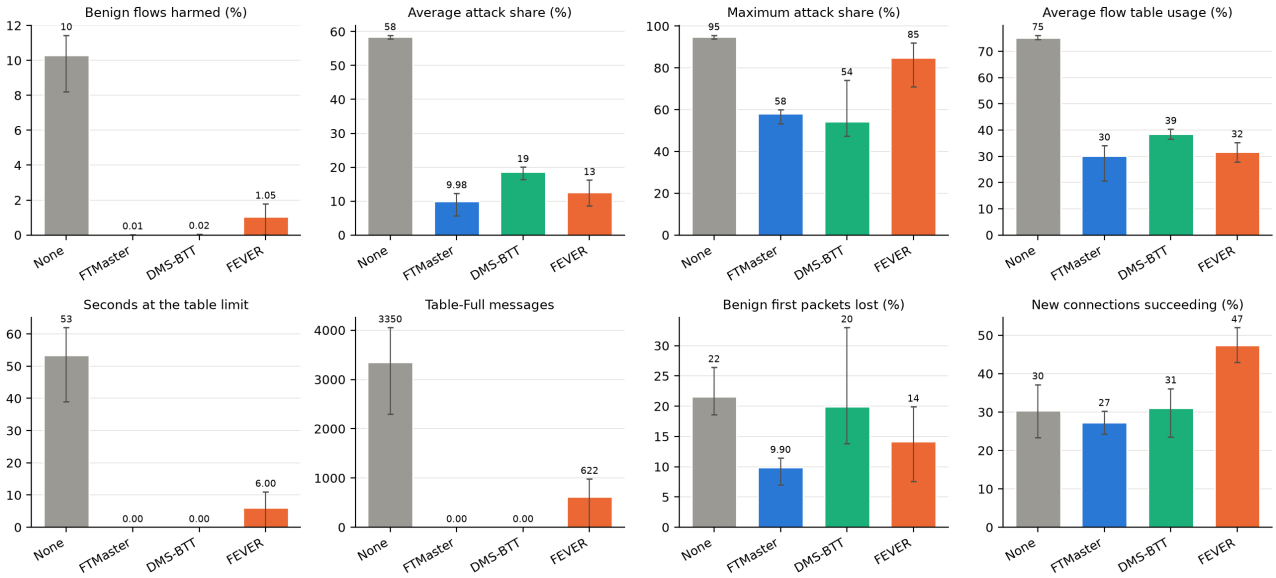


Fig. 1. Comparison of the defenses under A2 (five runs each; bar: mean, whisker: range).

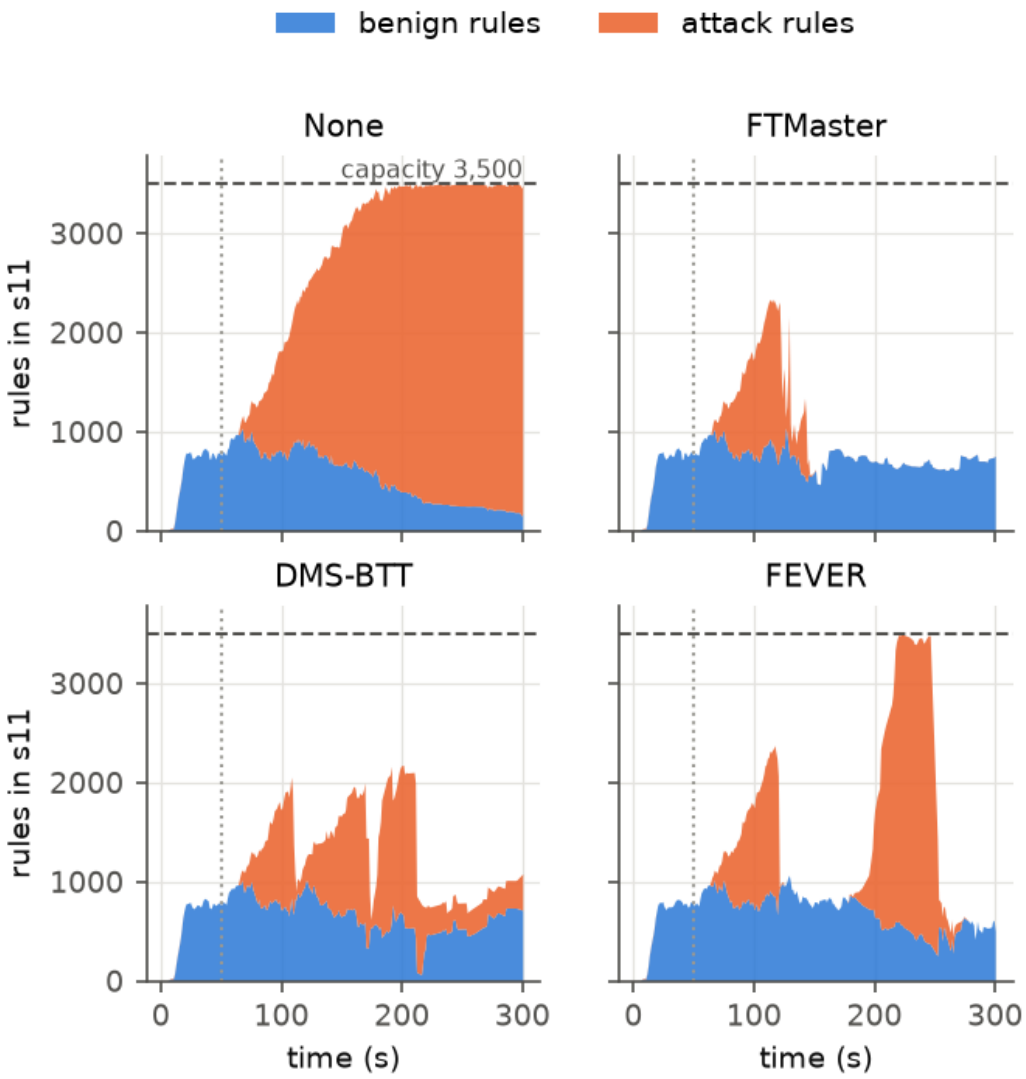


Fig. 2. Benign and attack rules in the flow table over time under A2.

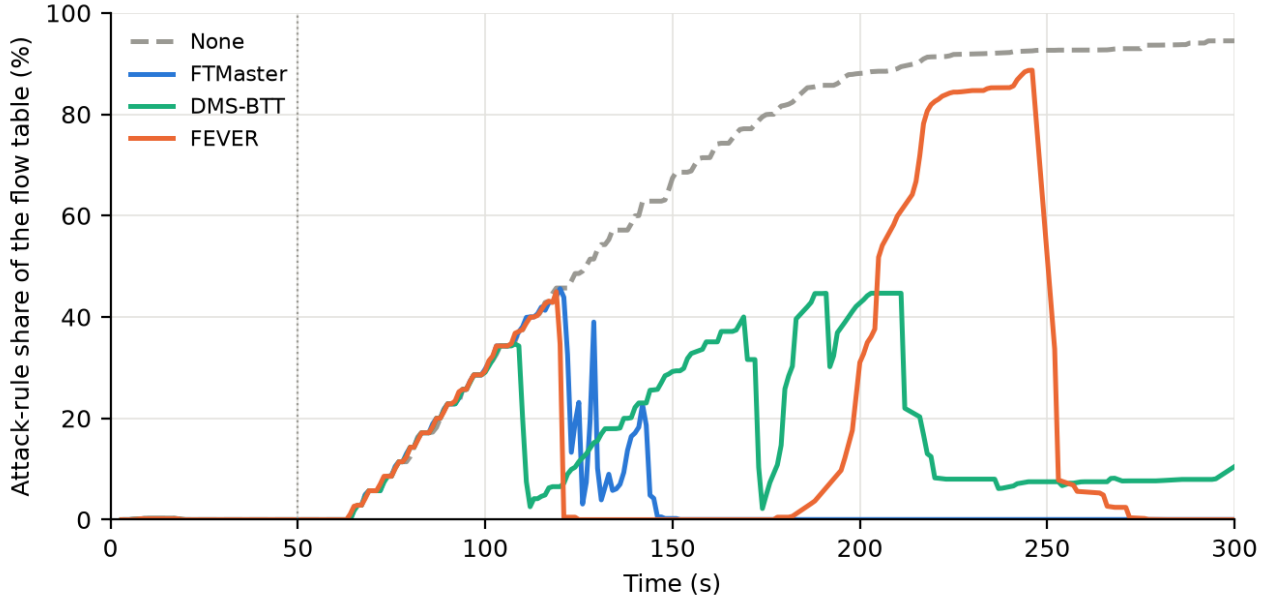


Fig. 3. Attack-rule share of the flow table over time under A2 (median of five runs).

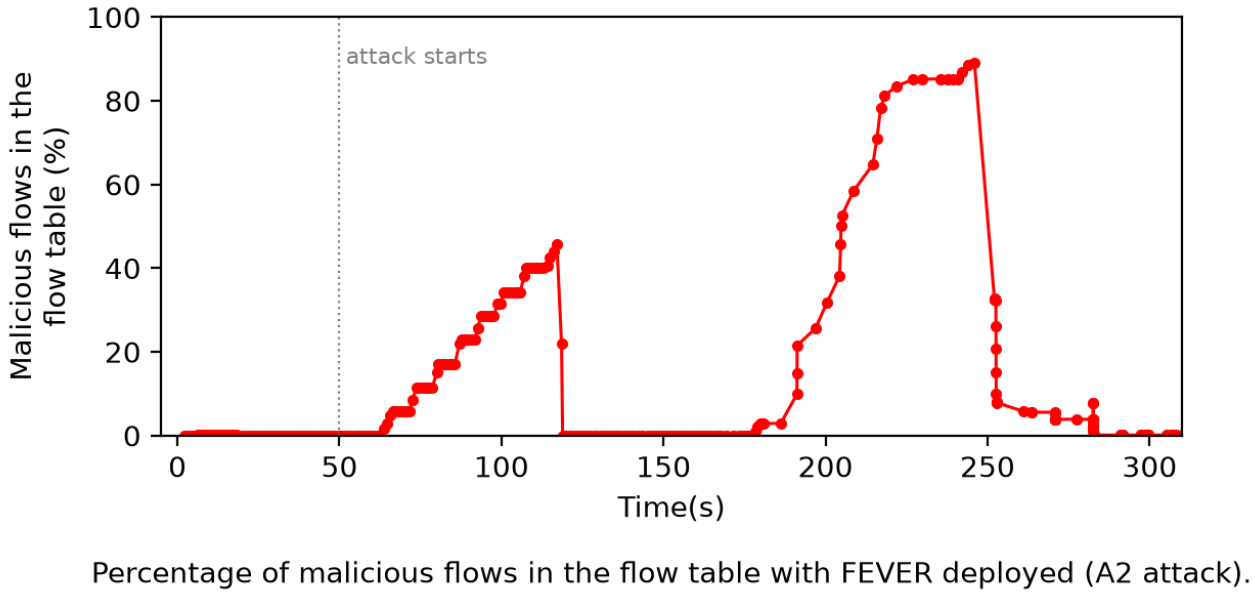


Fig. 4. Percentage of malicious flows in the flow table with FEVER under A2.

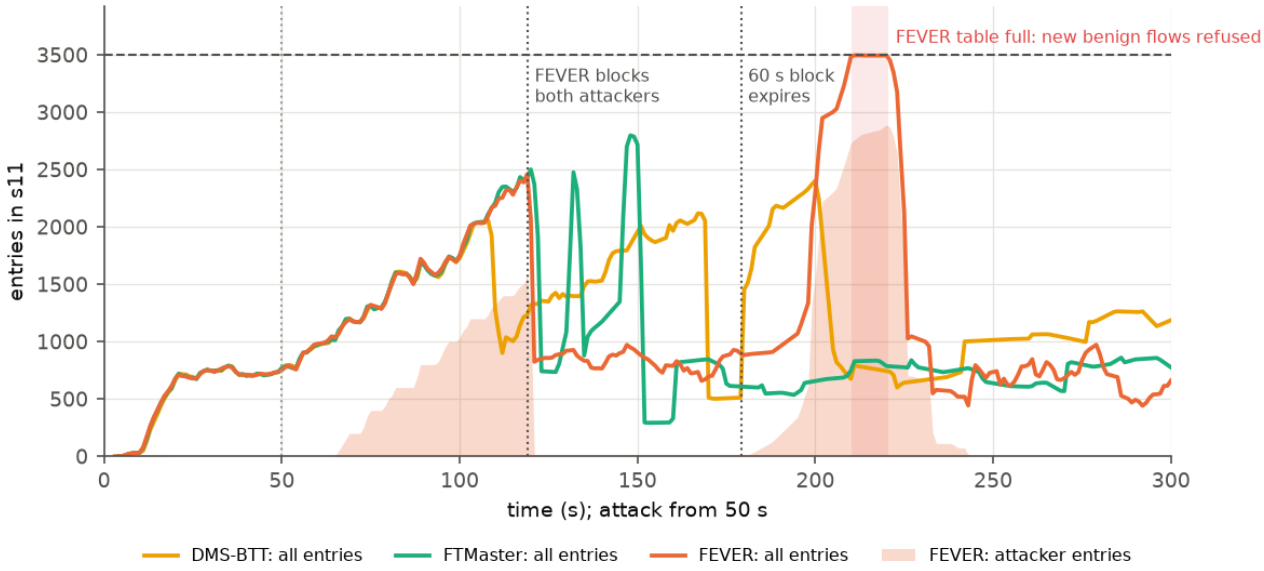
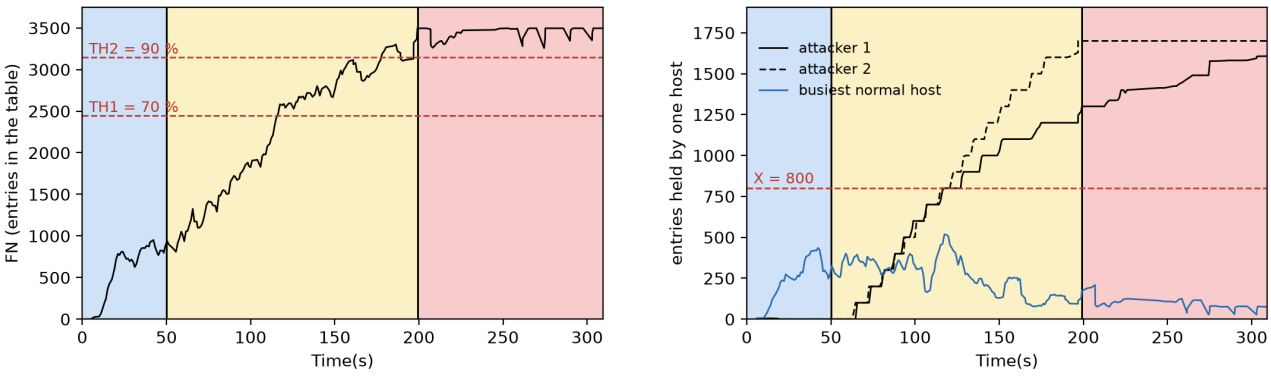


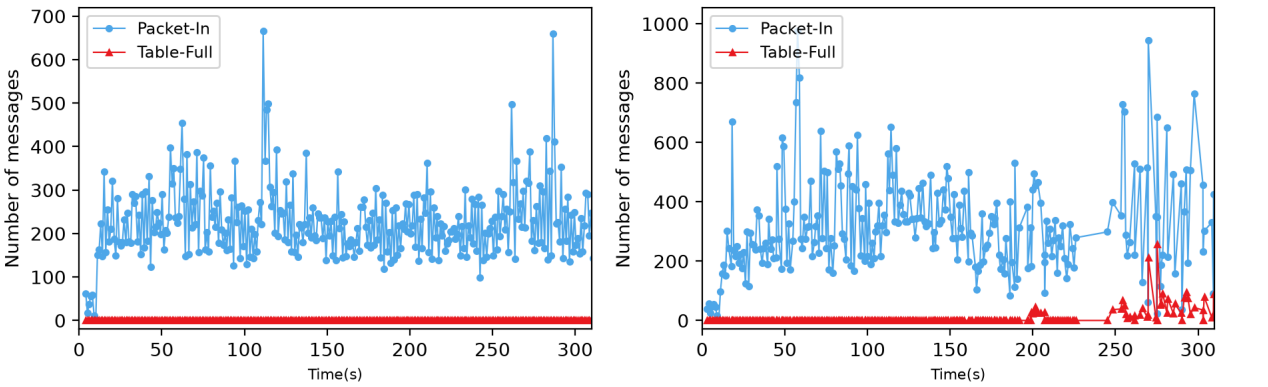
Fig. 5. Flow-table entries over time under A2.



(a) Distribution of FN before and after the attack.

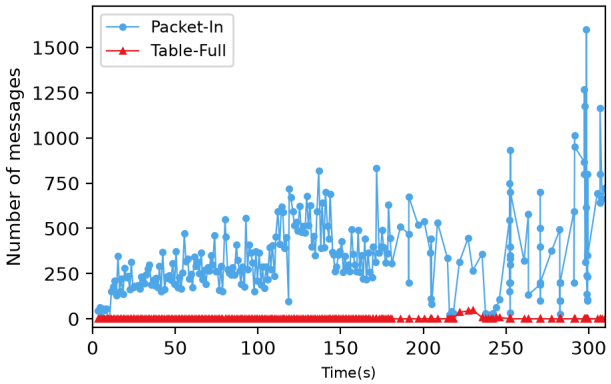
(b) Entries per host before and after the attack.

Fig. 6. Distribution of FN and of the entries held per host before and after the A2 attack.



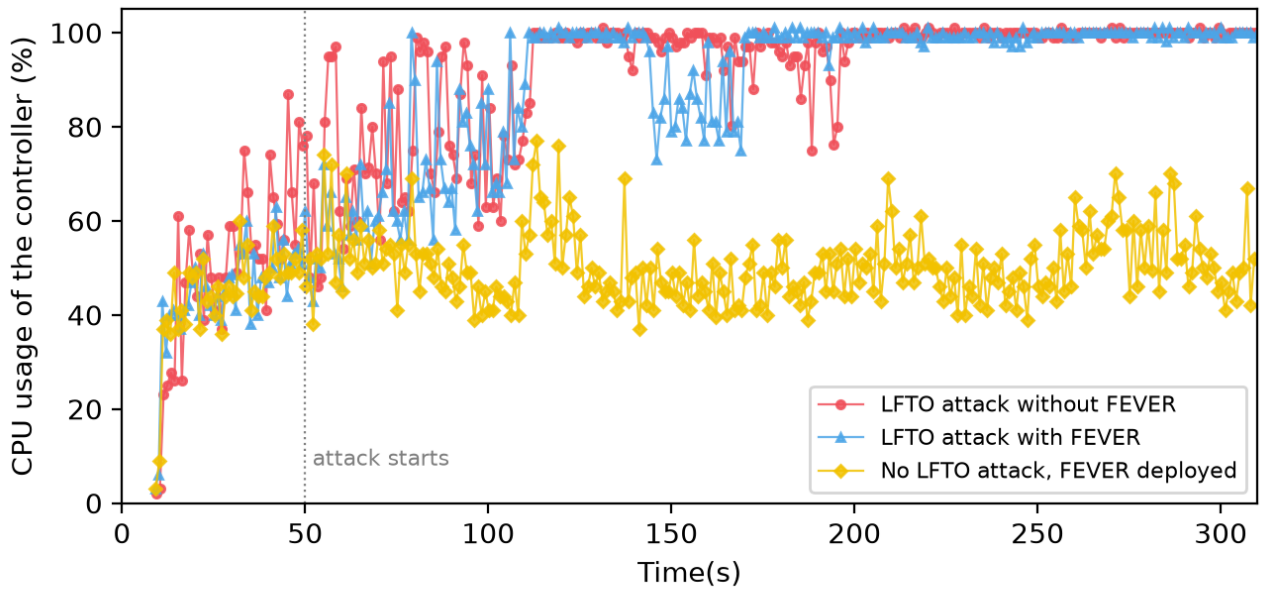
(a) No LFTO attack in the network.

(b) LFTO attack launched and FEVER not deployed.



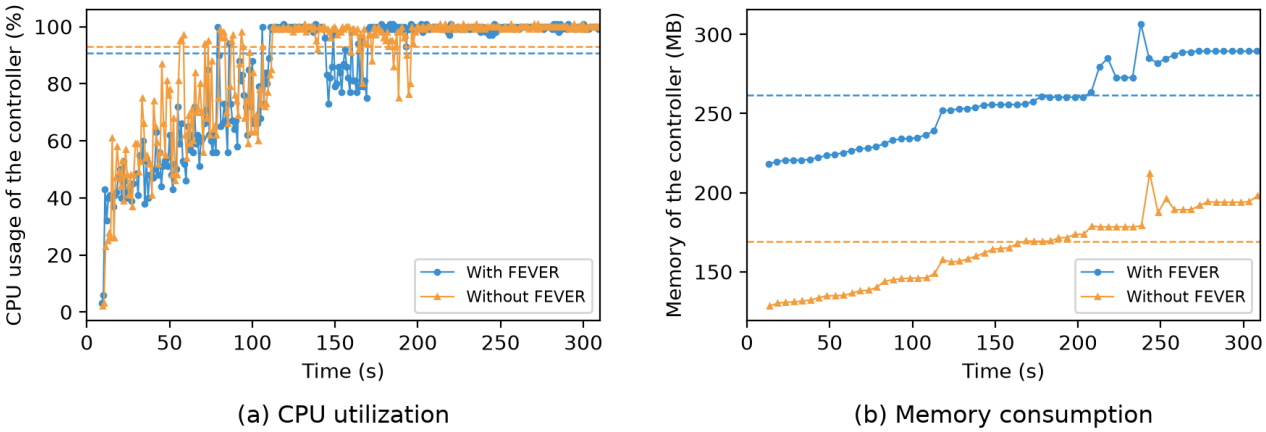
(c) LFTO attack launched and FEVER deployed.

Fig. 7. The quantity of Packet-In and Table-Full messages under A2.



CPU usage of the controller with and without FEVER (A2 attack).

Fig. 8. CPU usage of the controller under A2.



(a) CPU utilization (b) Memory consumption

Fig. 9. System overhead of deploying FEVER under A2.